MILITARY SPECIFICATION

TEST SET, BAROSWITCH, T4016A (T310), PACKAGE 1 OF 2

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of test set, entitled Test Set, Baroswitch, T4061A (T310) Package 1 of 2 which is utilized with the Nike Hercules warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1168	- Lot Numbering of Ammunition.
MIL-STD-1460	- Soldering of Electrical Connections and Printed Wiring Assemblies, Procedure For.

DRAWINGS

ORDNANCE CORPS

8786200	- M-5 Switch Assembly.
8786233	- Unit Assembly.
8804125	- Assembly.
8805913	- Test Set, Baroswitch, T4016A (T310), Package 1 of 2, Assembly, Packing, and Marking.
8808310	- Box, Packing, Ammunition, for Test Set, Baroswitch T4016A (T310), with Auxiliaries Package 1 of 1.
EL-8816847	- Equipment List.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The test set shall comply with all requirements specified on the drawing (dwg) 8805913, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Serialization. A six letter serial number shall be assigned to the item. Serial numbers required on these items shall be obtained by means of a letter addressed to: Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTENTION: SARPA-QA-N-A. The letter of request shall include:

a. Exact nomenclature of the item as specified on the applicable drawing.
b. Name of the contractor.
c. Contract number or procurement request order number.
d. Number of serial numbers required.

3.5 Electrical continuity. The test set shall exhibit electrical continuity when tested as specified in 4.4.1.

3.6 Leak test. The test set shall show no internal pressure change exceeding 85 mm Hg in less than 5.0 seconds due to leakage when tested as specified in 4.4.2.

3.7 Function. The test set shall function properly when tested as specified in 4.4.3.

3.8 Environmental tests (see 4.4.4).

3.8.1 Vibration. The test set shall show no evidence of damage and shall meet the requirements of 3.5 through 3.7 after subjection to a frequency range of 10 to 55 Hertz (Hz) at a double amplitude of 0.064 inch when tested as specified in 4.4.4.1.

3.8.2 Handling shock. The test set shall show no evidence of damage and shall meet the requirements of 3.5 through 3.7 after subjection to thirty-inch drops when tested as specified in 4.4.4.2.

3.8.3 High temperature. The test set shall show no evidence of damage and shall meet the requirements of 3.7 after subjection to a temperature of 160° ± 5°F when tested as specified in 4.4.4.3.

3.8.4 Low temperature. The test set shall show no evidence of damage and shall meet the requirements of 3.7 after subjection to a temperature of -65° ± 5°F when tested as specified in 4.4.4.4.

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3.8.5 Humidity. The test set shall show no evidence of deterioration or damage such as obliterated markings or distorted, cracked, chipped, peeled, swollen, or eroded components and shall meet the requirements of 3.7 when tested as specified in 4.4.4.5.

3.9 Soldering. All soldering shall comply with the requirements specified in MIL-STD-1460.

3.10 Workmanship.

3.10.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.10.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.10.3 Burrs. Parts shall be free from burrs.

3.10.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.10.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 3 complete assemblies (see dwg 8805913) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table I. Tests need not be conducted in order specified in Table I.

Table I
First article inspection and tests

Inspection or Test	Dest	Non-Dest
1. Classification of characteristics (see 4.3.2.1)		X

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2. Electrical continuity (see 3.5)		X
3. Leak test (see 3.6)		X
4. Function (see 3.7)		X
*5. Vibration (see 3.8.1)	X	
*6. Handling shock (see 3.8.2)	X	
7. High temperature (see 3.8.3)		X
8. Low temperature (see 3.8.4)		X
*9. Humidity (see 3.8.5)	X	

*One unit selected at random from the first article sample

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the pupose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristics table (see Table II and Table III), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Table II
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

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Table III
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Unit assembly (see dwg 8786233) .

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Improper location of electrical components	Visual
102.	Improper location of name plates	Visual
103.	Damaged wiring, such as abrasion, breaks, etc	Visual
104.	Pneumatic tube split or damaged	Visual
105.	Any component part loose or missing	Visual/Manual
106.	Hardware and fastener not installed properly and securely	Visual/Manual
107.	Loose wiring connection	Visual/Manual
108.	Fitting of tube assembly loose	Visual/Manual
109.	Improper soldering (see 3.9)	Visual
*110.	Height of pin(s) (pins of the same diameter must all appear to be within 1/16 inch in height relative to the adjacent pins)	Visual
**111.	Depth of socket(s) (female sockets of the same diameter must all appear to be within 1/16 inch in height relative to the adjacent sockets)	Visual

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Minor:
201. Plating damaged to the extent of exposing base metals.....Visual
202. Marking missing, misleading, or unidentifiable.....Visual
203. Evidence of poor workmanship (see 3.10).....Visual

4.3.2.1.2 Assembly, with cover removed prior to installing liner (see dwg 8804125).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Shock mount not fastened securely	Visual
	102. Shock mount not oriented properly	Visual
	103. Humidity indicator missing or not fastened securely	Visual/Manual
Minor:		
	201. Evidence of poor workmanship (see 3.10)	Visual

4.3.2.1.3 Assembly, with cover removed prior to installing unit assembly (see dwg 8804125).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Liner fastened insecurely	Visual/Manual
	102. Dehydrator fastened insecurely	Visual/Manual
Minor:		
	201. Evidence of poor workmanship (see 3.10)	Visual

4.3.2.1.4 Assembly prior to securing cover (see dwg 8805913)

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Unit assembly fastened improperly	Visual

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102. Damage to desiccant bag such as tear, hole, etc.....Visual
103. Desiccant bag not in place.....Visual
104. Sealing gasket containing foreign matter.....Visual
105. Tape not removed from dehydrator.....Visual
106. Contents of cover missing or improperly packed.....Visual
107. Serialization improper or missing (see 3.4).....Visual
108. Objective evidence relative to material missing or improper (see 3.1).....Visual

Minor:
201. Evidence of poor workmanship (see 3.10).....Visual

4.3.2.1.5 Assembly T4016A (T310) test set (Package 1 of 2) (see dwg 8805913).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Damaged case	Visual
	102. Improper fit of cover	Visual
	103. Seal not intact	Visual
	104. Marking, incorrect, illegible, missing	Visual
	105. Packing list missing	Visual
	106. Catches insecure or disengaged	Visual/Manual
	107. Handles loose or missing	Visual/Manual
Minor:		
	201. Evidence of poor workmanship (see 3.10)	Visual

4.3.2.1.6 Box, packing prior to sealing liner (see dwg 8808310).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Any loose splinters on inside of box	Visual

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102. Cushioning material damaged, missing or improperly located.....Visual
103. Case liner damaged or missing.....Visual
104. Test set not positioned properly.....Visual
105. Desiccant bags damaged or missing.....Visual
106. Humidity indicator damaged, missing or improperly located.....Visual

Minor:
201. Evidence of poor workmanship (see 3.10).....Visual

4.3.2.1.7 Box, packing prior to closing, sealing and strapping (see dwg 8808310).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Case liner damaged	Visual
	102. Case liner seal not intact	Visual
	103. Envelope or packing list missing or improperly located	Visual
	104. Insufficient filler material	Visual/Manual
Minor:		
	201. Evidence of poor workmanship (see 3.10)	Visual

4.3.2.1.8 Box, packing (see dwg 8808310).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Cover incorrectly sealed	Visual
	102. Box improperly sealed	Visual
	103. Strapping missing or incorrectly positioned	Visual
	104. Marking, missing, incorrect or unidentifiable	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.10)	Visual

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*In the event pin(s) do not appear to be within 1/16 inch in height relative to the adjacent pins, Gage (see 4.3.4) shall be used to determine the variation from the normal height. Reject any pin that varies by more than 0.063 inch max. from the normal pin height of the individual connector. The normal height of the individual connector pins shall be determined by calculating the average height from measurements taken of pins adjacent to the pin that appears to be defective. The forward metal edge of the connector shall be utilized as the reference surface of all measurements of pin heights.

**In the event socket inserts do not appear to be within 1/16 inch in depth relative to the adjacent socket, Gage (see 4.3.4) shall be used to determine the variation from the normal depth. Reject any insert that varies by more than 0.063 inch max. from the normal socket insert depth of the individual connector. The normal depth of the individual connector sockets shall be determined by calculating the average depth from measurements taken of sockets adjacent to the socket that appears to be defective. The forward edge of the connector insert shall be utilized as the reference surface for all measurements of socket depths.

4.3.2.1.9 Preparation for delivery (see 5).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Preservation, packing, and packaging not in accordance with (see dwg 8808310)	Visual
	102. Marking not in accordance with (see dwg 8808310)	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.10)	Visual

4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

	Test	Test Classification
a.	Electrical continuity	Critical
b.	Leak test	Major
c.	Function	Critical
d.	High temperature	Major
e.	Low temperature	Major

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f. Vibration	Major
g. Humidity	Major

Of these tests, g and f are considered destructive and samples so tested shall not be returned to the lot. Destructive samples are over and above the deliverable quantity of items required by the contract.

4.3.3.1 Electrical continuity, leak test, and function. Each T4016 (T310) shall be tested in accordance with 4.4.1, 4.4.2 and 4.4.3. If any item fails to comply with 3.5 through 3.7, the item shall be rejected and removed from the lot.

4.3.3.2 High temperature, low temperature, vibration, and humidity. One sample T4016A (T310) test set shall be selected at random by the Government inspector from each 25 consecutively produced units, and shall be tested in accordance with 4.4.4.1, and 4.4.4.3 through 4.4.4.5. If the sample fails to comply with 3.8.1 and 3.8.3 through 3.8.5, all units which have not been accepted by the Government shall not be accepted until the cause of failure has been determined and corrected and retested, and the units withheld have also been corrected and retested.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Major characteristics and test. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (see EL-8816847). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Electrical continuity. The continuity shall be checked by approved test method (see 4.3.4.1).

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4.4.2 Leak test.

4.4.2.1 Test setup. The T2037's shall be properly connected in their respective positions in the test set. Care should be taken to tighten barohousing, hold-down screws uniformly to assure seating. The ambient atmospheric pressure or simulated equivalent shall not be less then 609 mm Hg absolute (6,000 feet maximum altitude) during the test. A closed-system pressure-measuring device (not part of test set) having an effective volume not in excess of 10 cubic inches shall be connected to the hose part. The inlet-outlet ports on top of both desiccant chambers shall be sealed during the test. The sealing of these ports should be made after the pneumatic system has been evacuated below 140 mm Hg absolute.

4.4.2.2 Test. With the internal pressure adjusted to 140 mm Hg absolute or less, the internal pressure change due to leakage shall not exceed 85 mm Hg in less than 5 seconds. During this time interval, the test set power shall be off. Upon completion of this test, remove the seals from the inlet-outlet ports on the desiccant chambers.

4.4.3 Function. The setup for these tests shall be as follows: Connect the T4016A (T310) test set both electrically (with its W7H cable) and pneumatically (with hose provided) to a functional M-5 (refer to drawing 8786200) in which all four baro elements are known to close within + 2.5 mm Hg of any nominal setting that may be indicated on the B-set indicator: connect the pressure-measuring device (see 4.3.4.1) to the pneumatic system; connect ground strap to a good earth ground; connect cable W3H to a 117 volts ac, 60 Hz power source.

4.4.3.1 Step I. Starting with all switches in their OFF or normal position, turn ac power switch ON. Blower shall start operating. The ac power lamp shall light. Press the "Ready", "Accept", and "Reject" lamps.

4.4.3.2 Step II. Starting with all switches in their OFF or normal position, turn ac power switch ON and turn S1 switch to position 2. The closed-system pressure measuring indicator shall immediately indicate dive. This indicator shall reverse direction indicating a change to climb within 20 seconds. Whithin sixty seconds after the change to climb indication, the "Ready" lamp shall have lighted and remain lighted. Under these load conditions, 28.5 $\pm$ 2 volts dc shall be observed on pin E of J8 and 27 $\pm$ 3 volts ac, 400 $\pm$ 40 Hertz(Hz) shall be observed between pins B and C of J8.

4.4.3.3 Step III. Turn S1 to position 1. The T4016A, (T310), test set shall cycle and the closed-system pressure measuring device shall indicate a dive and climb alternately and the "Ready" lamp shall flash momentarily at the top of each climb. The dive to climb change shall occur between 818 and 846 mm Hg absolute. The climb to dive change shall occur between 187.4 and 137.4 mm Hg absolute.

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4.4.3.4 Step IV. Turn S1 to position 2 (Note: if the cycle is at the top of its climb as indicated by the "Ready" lamp being on momentarily, wait until lamp goes out before changing S1). Set the B-set indicator to 1,000 feet. When the "Ready" lamp has lighted and the 1,000 foot setting is made, push the"Test " switch. (Note: "Ready" lamp will remain lighted until just after the "Test" switch is pushed). The closed-system pressure measuring indicator shall indicate a change from climb to dive, the "Ready" lamp shall go out, and before the closed-system pressure measuring indicator again indicated climb the green "Accept" lamp shall have lighted. Repeat the above with S1 set at position 2 and B-set indicating 5,000 feet; with S1 set at position 4 and B-set indicating 15,000 feet; and with S1 set at position 5 and B-set indicating 25,000 feet.

4.4.3.5 Step V. Set B-set to 20,000 feet, S1 remains in position 5. When the "Ready" lamp lights, push the"Test" switch, the closed-system pressure measuring indicator shall indicate a change from climb to dive and during the dive sequence, the "Ready" lamp shall go out. Before the climb indication appears again, the red "Reject" lamp shall light and remain lighted, and the dive sequence shall stop. Turn ac power off on proceeding to next step.

4.4.3.6 Step VI. Turn S1 to position 4. "Reject" lamp shall go out, the dive-climb cycle shall commence. The 20,000 feet setting shall not be changed. When the "Ready" lamp again lights, push the "Test" switch. The closed-system pressure measuring indicator shall indicate a change from climb to dive and during the dive sequence the "Ready" lamp shall go out. Before climb indication appears again, the red "Reject" lamp shall light and remain lighted, and the dive sequence shall stop. Extinguish the "Reject" lamp by momentarily turning off ac power. The dive-climb sequence will proceed when ac power is turned on again.

4.4.3.7 Calibration circuit check. With S1 in any position other than OFF, rotate S6 to its momentary-ON position and hold it in this position until the B-Set indicator stops at zero. The zeros shall be clearly visible. While still holding S6 in this momentary position, operate the "Cal" switch. The B-Set shall drive to some position indication greater than 100 feet. Release S6 and "Cal" switches. The B-Set shall restore to the 20,000 feet indication.

Note: After completing above tests, drive B-Set to any desired setting and restore all switches to their OFF or normal position.

4.4.4 Environmental testing (see 3.8).

4.4.4.1 Vibration. The T4016A (T310) test set, with cover attached shall be secured to a vibration table and vibrated for a 45 minute period in each of three mutually perpendicular direction through a frequency range of 10 to 55 Hz at a double amplitude of 0.064 inch.

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The frequency shall be varied uniformly from 10 to 55 to 10 Hz making approximately one cycle per minute. After completion of this test, the test set shall be tested as specified in 4.4.3 using the cables and hose contained in the cover for all connections.

4.4.4.2 Handling shock. The T4016A (T310) test set with cover attached shall be dropped on a concrete surface from a height of 30 inches. (The height of 30 inches refers to the distance form the concrete to the nearest face of the container when suspended prior to the fall). The fall shall be a free fall in that no ropes or other suspending media are attached to the equipmemt during the fall. One drop shall be made for each face or side of the T310 test set. After completion of this test the cover shall be removed and the test set shall be tested as specified in 4.4.1, 4.4.2, and 4.4.3 using the cables and hose contained in the cover for all connections.

4.4.4.3 High temperature. The T4016A (T310) test set with cover attached shall be placed in a test chamber and stabilized (see 6.4) at a temperature of 160 ± 5°F with a maximum relative humidity of 15 percent for a period of ten hours. It shall then be reduced to room temperature (75 ± 10°F) and tested as specified in 4.4.3 using the cables and hose contained in the cover for all connections.

4.4.4.4 Low temperature. The T4016A (T310) test set, with cover attached, shall be placed in a test chamber and the temperature reduced and stabilized at -65 ± 5°F. The chamber temperature shall be held for a minimum of two hours. The temperature shall then be raised to ambient room temperature (75 ± 5°F) without the use of any forced means and the test set shall be tested as specified in 4.4.3 using the cables and hose contained in the cover for all connections.

4.4.4.5 Humidity. The T4016A (T310) test set in its case with cover removed, with all connections made, and in normal position shall be placed in a test chamber. The humidity shall be maintained at 90 to 98 percent relative humidity throughout the following cycle. The temperature shall be raised to 130°F within four hours and maintained at this point for eight hours. The temperature shall then be reduced to 86°F within four hours and maintained at this point for 21 hours. Between the 8th and 12th hours after stabilization at 86°F, the T310 test set shall be tested as specified in 4.4.3. The temperature shall be reduced to 68°F within one hour and maintained at this point for four hours. The temperature shall then be raised to 86°F within one hour and maintained at this point for five hours. At the conclusion of this test the test set shall be tested as specified in 4.4.3.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The test set shall be packaged in accordance with dwg 8808310 using the specified moisture barrier.

5.1.2 Level C. The test set shall be packaged in accordance with dwg 8808310 except that the moisture barrier may be omitted.

5.2 Packing.

5.2.1 Level A and C. The test set shall be packed in accordance with dwg 8808310.

5.3 Marking. Marking shall be as specified in accordance with dwg 8808310.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.

b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-T.

6.4 Temperature stabilization. The temperature is considered stabilized when the greatest peaks (both highest and lowest) continually observed over 5 minutes min. do not exceed the nominal temperature plus tolerance specified.

6.5 Supersession data. This specification supersedes the requirements of Purchase Description X-PA-PD-1181 dated 4 November 1959.

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No. 1135-A184

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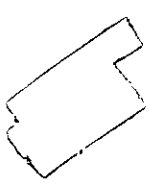
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

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8/4